Scanning Magnetoresistance Microscopy of Tape Writers and Media

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We report on the development of a scanning magnetoresistance microscope optimized to study the static and dynamic magnetic fields produced by tape write transducers and recorded tape media samples. We demonstrate the spatial and temporal resolution of the microscope by imaging recorded patterns on a perpendicularly oriented tape media sample and the static and dynamic fields produced by a tape write transducer.

Index Terms—**magnetic recording, magnetic tape, scanning magnetoresistance microscope, write transducer**

I. Introduction

DIGITAL magnetic tape technology has a long history over which it has maintained its importance in the storage hierarchy through continuous scaling of its areal recording density and data rate [1]. The continued scaling of these metrics will be critical for tape technology to continue to maintain its relevance in the future [2]. The high resolution characterization of magnetic field distributions will be essential for the development and optimization of future magnetic recording media and write transducers. As transducer dimensions and bit sizes continue to decrease while data rates simultaneously increase, measurement techniques are required that provide both a high spatial and high temporal resolution of the magnetic fields produced under realistic operating conditions. Magnetic force microscopy (MFM) can achieve very high spatial resolution but has a limited temporal resolution. To address this challenge, we have developed a scanning magnetoresistance microscope (SMRM) to study the magnetic fields produced by recorded tape media samples and tape write heads under typical recording conditions. Our instrument is similar to previously described SMRMs that were developed for HDD research, but has been customized for application to magnetic tape recording [3], [4].

II. Experimental Setup

A schematic diagram of the SMRM is shown in Fig. 1. We tested the instrument using two types of shielded giant magneto-resistive (GMR) hard disk drive (HDD) read sensors. The HDD sliders were mounted on a gimbal assembly, which remains in a fixed position during operation while the sample is raster scanned below the sensor. The sample—either a tape write head module or a section of magnetic tape—is mounted on a piezo-driven X–Y–Z scan table with closed-loop position control. Coarse alignment and angular adjustment of the sensor relative to the sample are achieved via manual translation and tilt stages. For tape write transducer characterization, the write head module is connected via flex cable to the electronics card of a commercial tape drive that is equipped with write drivers capable of generating arbitrary write current waveforms.

Two scan modes are supported: (i) contact mode, for imaging written magnetic patterns on tape, where the sample surface is brought into controlled contact with the read head; and

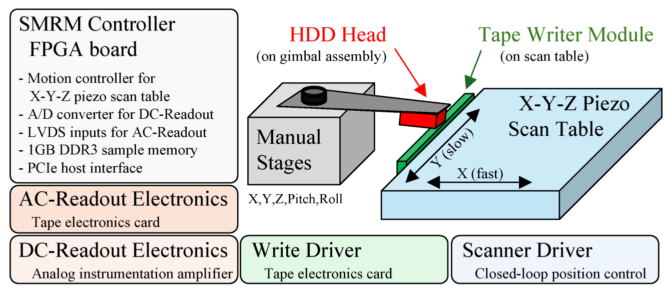

Fig. 1. Architecture of the SMRM configured for imaging a tape write head module. To image tape media, the writer is replaced by a tape sample holder.

(ii) non-contact mode, for write transducer characterization, where the Z-axis maintains a defined sensor–device separation. Signal acquisition is implemented via two complementary read-back paths. A DC-coupled path, consisting of a low-noise instrumentation amplifier with a DC compensation circuit and a 12-bit A/D converter (ADC), is optimized for static or quasi-static field mapping. In parallel, an AC-coupled path based on an IBM[1] TS1140 tape electronics card with dedicated analog front/back-end ASICs and high-speed ADCs enables time-resolved field measurements. A custom built controller based on an FPGA coordinates the synchronized operation of scan trajectory, signal acquisition and data buffering, and transferring of the measurement results to a host PC. This modular system decouples mechanical scanning, sensing, acquisition, and control, enabling flexible operation and future extensions.

III. Results

To test the spatial resolution of our microscope we imaged servo and data patterns written on an IBM TS1160-JE tape media sample which has a perpendicular magnetic orientation using the DC-coupled readout electronics and with a $\approx$140 nm wide read head in contact with the tape sample. Figure 2 shows captured SMRM images depicting three bursts of servo stripes of the factory pre-formatted timing-based servo pattern (TBS), and the edge of a data track recorded with a periodic $8T$ pattern with bit length $T = 47\,\mathrm{nm}$.

Figure 3 highlights the result of SMRM imaging an IBM TS1150 tape write transducer driven by a static write current of

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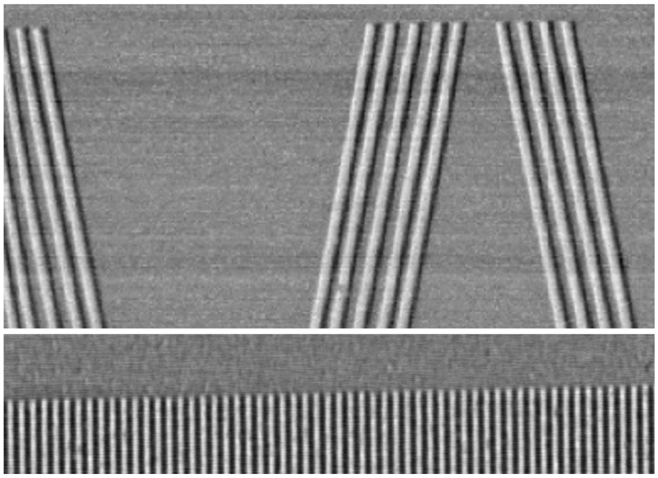

Fig. 2. SMRM images of an IBM TS1160-JE tape. Top: $90\,\mu\text{m} \times 45\,\mu\text{m}$ scan showing three servo-stripe bursts from the timing-based servo pattern. Bottom: Recorded periodic data pattern ($8T$); scan size $42\,\mu\text{m} \times 9\,\mu\text{m}$.

$27\,\text{mA}$. Imaging is done in non-contact mode, using the DC-coupled readout path with a $\approx$100 nm wide reader, measuring an area of $16\,\mu\text{m} \times 16\,\mu\text{m}$ around the write gap located between the P1 and P2 writer poles. Each data point represents an $8\times$ averaged sensor output voltage measurement, which is proportional to the MR head's resistance change.

Dynamic write field measurements of the tape writer using the AC-coupled path are shown in Fig. 4. A repeating pseudo-random binary sequence (PRBS) of length N = 63 generated using the polynomial $x^6 + x^5 + 1$ is used to modulate a bipolar write current waveform with clock frequency $f_\text{W} = 42\,\text{MHz}$. At each scan point (pixel), the system captures about 100 periods of the PRBS-63 pattern at a sampling rate of $f_\text{R} = 1.25 f_\text{W}$ and fixed analog gain. The captured signal is processed offline to compute signal RMS. Each signal capture is divided into single PRBS-63 periods which are then averaged to produce a low-noise signal waveform. Channel responses calculated by means of the known PRBS input are used to estimate the signal polarity and phase.

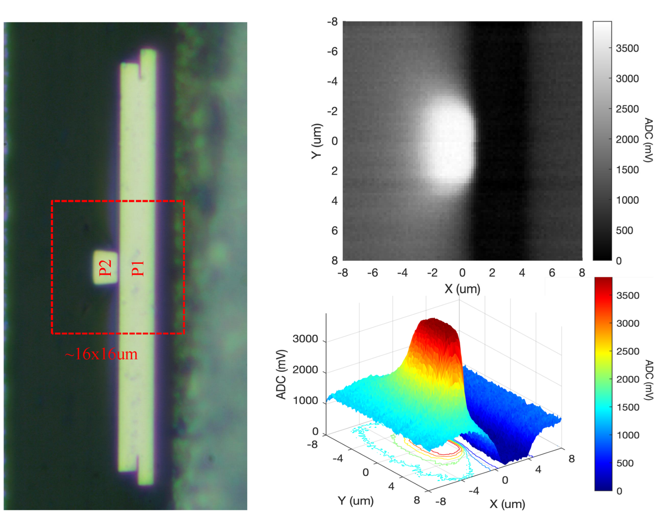

Fig. 3. Left: Microscope image of IBM TS1150 writer. Right: $16\,\mu\text{m} \times 16\,\mu\text{m}$ SMRM image of the write head field when driven at a fixed current of $27\,\text{mA}$.

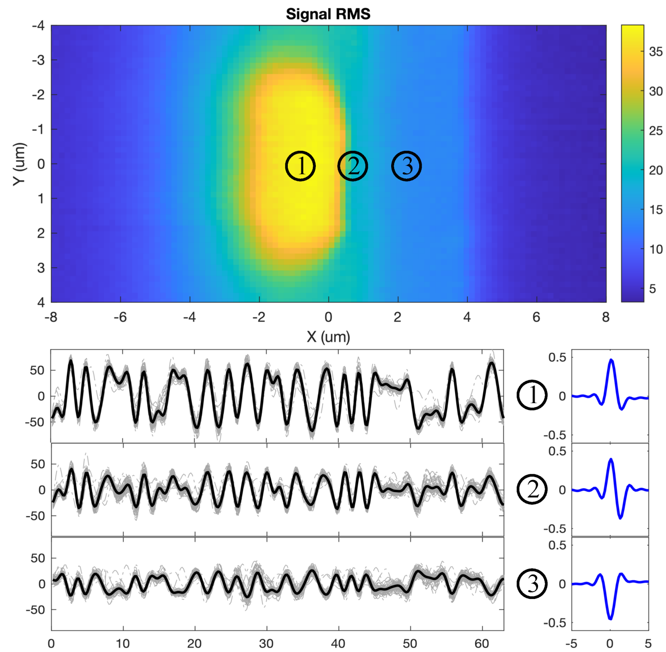

Fig. 4. Fixed-gain SMRM AC-readout results of the IBM TS1150 writer driven with a bipolar PRBS-63 current waveform. Top: Image representing the readback signal RMS at each location. Bottom left: Overlay of readback signals showing one PRBS-63 period and averaged signal (thick line), captured at locations 1, 2, and 3. Bottom right: Corresponding channel responses.

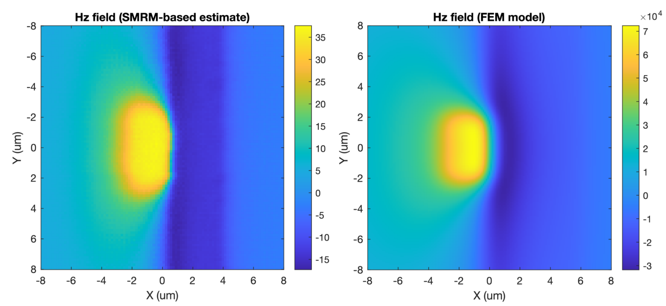

Fig. 5. Estimated write field component H_z from SMRM data (left image) vs. FEM calculation $1\,\mu\text{m}$ above the pole surface (right image, A/m).

We compared the write field component H_z estimated from SMRM data to the results of finite element model (FEM) calculations and found qualitatively good agreement with the field calculated $1\,\mu\text{m}$ above the writer, as shown in Fig. 5. In the future we will extend this work to study state-of-the-art and prototype tape write transducers.

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